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Pil E.A.
Academician RANH, Professor, Dr. Sc. Eng., St. Petersburg

ANALYSIS OF 3D VEU GRAPHS WITH NEGATIVE VALUES OF VARIABLES WITH CALCULATION OF X6

Last year, the Nobel Prize in Economics was awarded to three American economists who significantly improved the understanding of the role of banks in the economy, especially during financial crises, by giving money to banks and the population to restore purchasing activity in the country. Here it is worth recalling at once that in a number of countries, such as Sweden, Switzerland, Denmark and Japan, banks issued some customers with a negative rate, i.e. customers returned an amount less than they received [1, 2, 3].

This article considers the issue of calculating the values of the lower critical volume of the Veu spherical economic shell and constructing 3D drawings when the values of three variables change, i.e. Veu (GDPeu) = f(X1, X2, X3, X4, X5, X6). In some cases, when the Veu values had small minima and maxima compared to the last calculated Veu value, then 3D drawings were built from the calculation so that the reader would understand more clearly how these changes occur.

Here, Figure 1 shows a 3D graph of Veu, when the values of the variables were as follows: X1= X1= X3= 1, X4=0.99, X5=-2.1...-3.0 and 3.13.13.85. Here Veu has negative values and decreases by 22.1 times. In the next Fig. 2 the depicted 3D graph Veu with variables X1= X1= X3= 1, X4=-0.099 ...-0.99, X5=-2.1...-3.0, X6= 2.67...3.13 has a minimum of -173.48 at point 9.

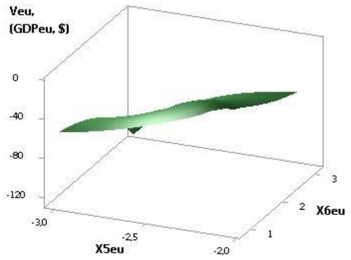

Fig. 1. 3D of Veu (GDPeu)

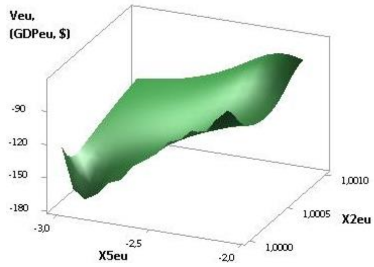

Fig. 2. 3D of Veu (GDPeu)

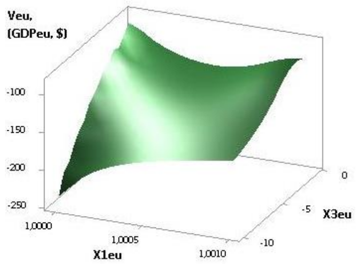

Fig. 3. 3D of Veu (GDPeu)

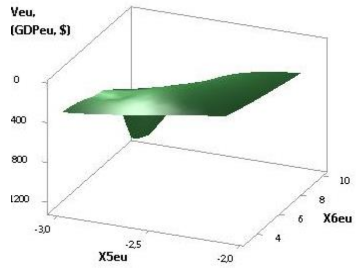

Fig. 4. 3D of Veu (GDPeu)

The next two Figures 3 and 4 show 3D graphs of Veu, when the variables were X1= X1= 1, X3=-1...-10, X4=0.1...0.99, X5=-2.1...-3.0, X6= 3.16...4.34 and X1= 1, X2= X3=-1...-10, X4=-0.1...-0.99, X5= -2.1...-3.0, X6= 0.92...11.79. As we can see, the constructed 3D graph of Veu in Fig. 3 decreases by 2.69 times and in Fig. 4 it decreases by 14.29 times.

The calculated values for the 3D Veu graph in Figure 5 for variables X1= X2= X3= -1...-10, X4=-0.1...-0.99, X5= -2.1...-3.0 and X6= 2.77...2.23 increase by 1.47 times. From the next Figure 6 it can be seen that for variables X1= X2= X3= -1...-10, X4=-0.1...-0.99, X5= 0 and X6= 2.77.2.03, the Veu values decrease by 1.86 times.

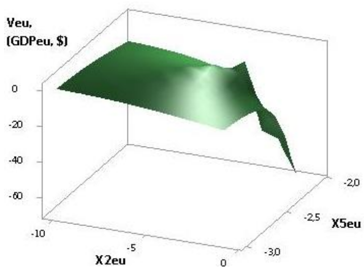

Fig. 5. 3D of Veu (GDPeu)

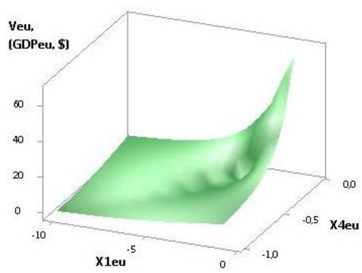

Fig. 6. 3D of Veu (GDPeu)

Figures 7 and 8 were plotted at X1= -1...-10, X2= X3= 1, X4=0.99, X5= 2.1, X6= 4.12...9.52 and X1= X2= -1...-10, X3= 1, X4=0.99, X5= 2.1, X6= 4.12...90.52. Here in Fig. 7 the Veu values increase by 5.33 times and in Fig. 8 it increases by 486.86 times.

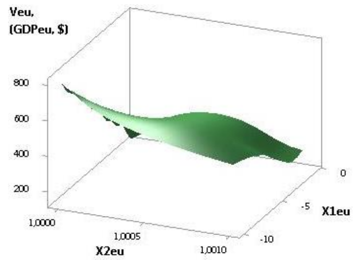

Fig. 7. 3D of Veu (GDPeu)

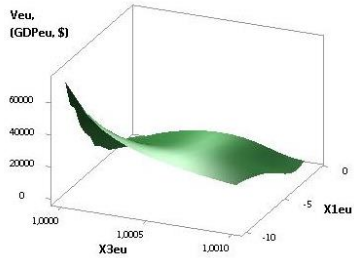

Fig. 8. 3D of Veu (GDPeu)

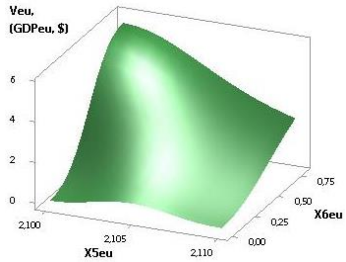

Fig. 9. 3D of Veu (GDPeu)

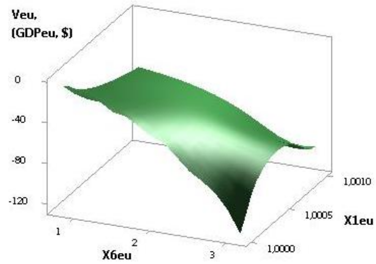

Fig. 10. 3D of Veu (GDPeu)

The next Figures 9 and 10 were plotted at X1= X2= X3= -1...-10, X4= 0.99, X5= 2.1, X6= 0.8 and X1= X2= X3= -1...-10, X4= 0.99, X5= 2.1, X6= 0.8. Here in Figure 9, the Veu variable has the only solution is 5.58 at point 1 and in Figure 10 has a single solution of 5.58 at point 1.

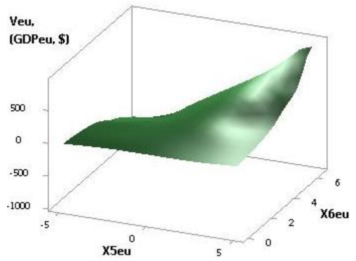

Fig. 11. 3D of Veu (GDPeu)

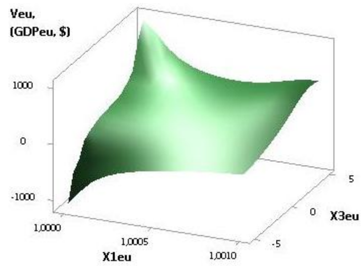

Fig. 12. 3D of Veu (GDPeu)

The last figures 11 and 12 were plotted at X1= X2= X3= 1, X4=0.99, X5= 5.0...-5.0, X6= 6.47...6.47 and X1= X2= 1, X3= X5= 5.0...-5.0, X4=0.99, X6= 6.95...7.19. In these examples, in Figure 11, the variable Veu is reduced by 9.96 times and in Figure 12 it is reduced by 147.4 times.

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